

ITA0611 – MACHINE LEARNING

ASSIGNMENT REPORT

Design and Development of an Intelligent Early-Warning System for Crop Disease and Yield Risk Prediction Using Machine Learning

Item	Details
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Course Code	ITA0611 – Machine Learning
CO Mapping	CO6, CO7, CO8
Bloom's Level	L5 – Evaluate; L6 – Create
SDG Mapping	SDG 2 – Zero Hunger; SDG 9 – Industry, Innovation and Infrastructure; SDG 13 – Climate Action

1. Problem Formulation

The system treats crop disease risk as a three-class classification problem (Low, Medium, High) and crop yield as a continuous regression problem measured in tonnes per hectare. Inputs include crop type, temperature, rainfall, humidity, soil moisture, N/P/K nutrients, soil pH and visible symptoms. The objective is to provide early warnings before severe disease or yield loss occurs.

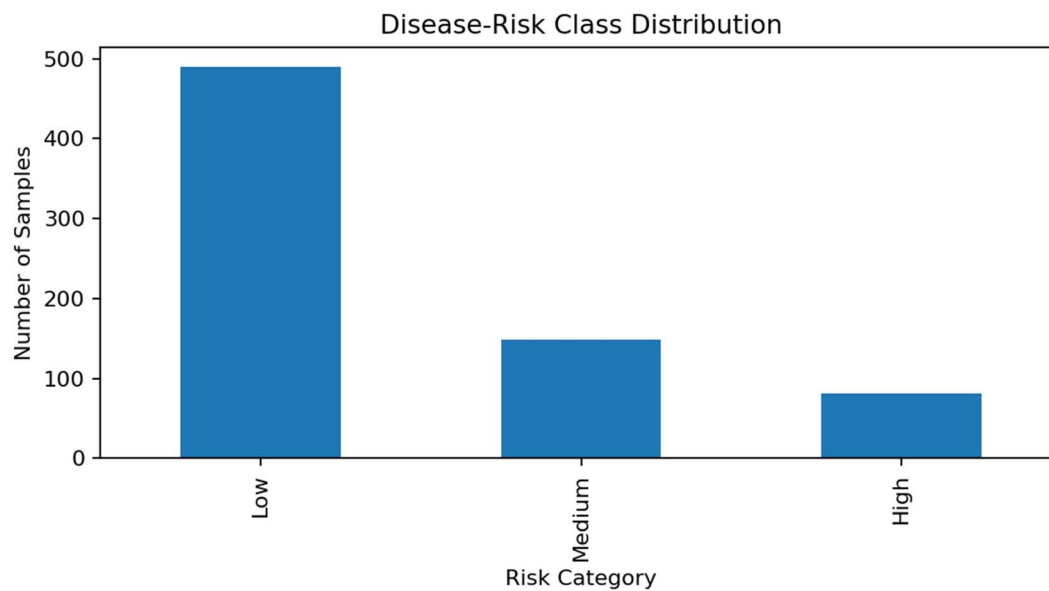
Expected Outcomes

- Clean and statistically analyse agricultural data using Pandas and NumPy.
- Represent disease-risk prediction as concept learning and perform Candidate-Elimination/Version-Space analysis.
- Develop Decision Tree, Bayesian, KNN, LWR and MLP models and examine their behaviour.
- Use a Genetic Algorithm for feature-selection/model optimisation.
- Compare predictive performance and create an integrated farmer-facing warning with recommendations.

2. Dataset Description and Preprocessing

The demonstration dataset contains 720 cleaned observations after duplicate removal. A small number of missing numerical values were introduced so that cleaning could be demonstrated. Numerical missing values were replaced with the median of the corresponding feature; categorical values were retained.

Feature	Mean	Std	Min	Max
Temperature_C	25.22	5.22	8.34	41.59
Rainfall_mm	80.09	32.57	10.0	170.92
Humidity_pct	71.93	12.84	29.67	100.0
SoilMoisture_pct	67.11	16.94	15.5	100.0
Nitrogen	86.11	19.62	35.01	148.37
Phosphorus	38.15	9.81	8.45	71.9
Potassium	34.22	10.8	5.0	66.24
Soil_pH	6.28	0.67	4.5	8.28
LeafSpot	0.18	0.39	0.0	1.0
Wilting	0.08	0.27	0.0	1.0
Yellowing	0.09	0.29	0.0	1.0
SymptomScore	0.36	0.56	0.0	2.0
Yield_t_ha	3.85	0.69	2.36	5.15



The class-distribution plot confirms that all three disease-risk categories are represented. Standardisation was used for distance-based KNN and LWR, while one-hot encoding was used for crop type in the tree, Bayesian and MLP pipelines.

3. Workflow Diagram

DATA COLLECTION



DATA CLEANING → missing-value treatment → duplicate check



EXPLORATORY ANALYSIS → statistics → correlations → visualisations



FEATURE ENGINEERING / ENCODING / SCALING



Concept Learning | Decision Tree | Bayesian | KNN



LWR for yield prediction → MLP + Back Propagation → GA Feature Selection



MODEL EVALUATION & COMPARISON



INTEGRATED EARLY-WARNING SYSTEM

↓

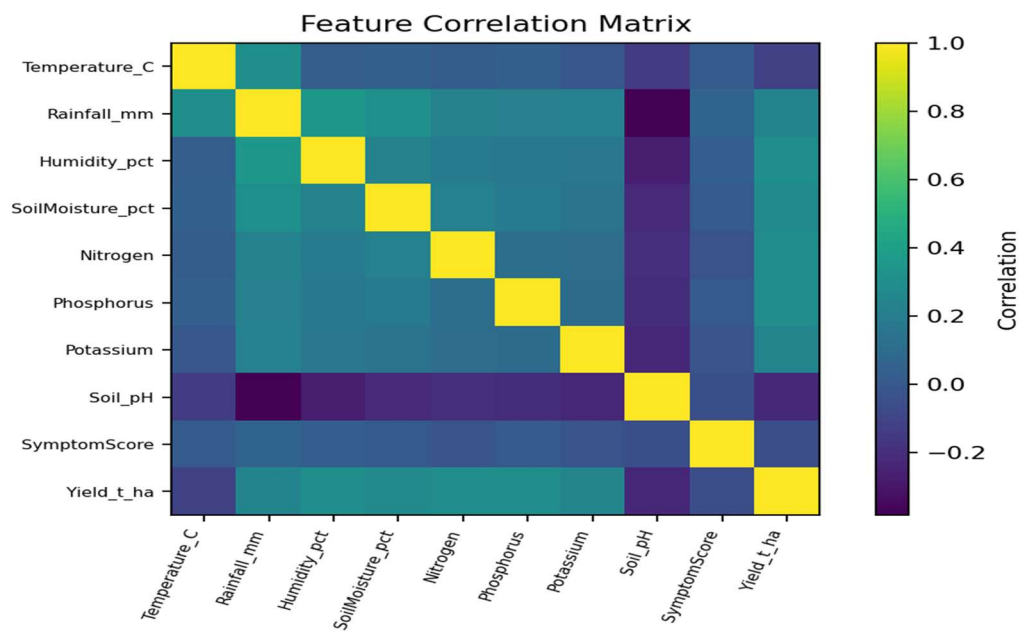
RISK + YIELD + RECOMMENDATIONS

4. Question 1 – Data Collection and Preprocessing

The variables were selected directly from the problem statement: temperature, rainfall, humidity, soil moisture, soil nutrients, crop type, disease symptoms and historical/target yield. Pandas was used for tabular cleaning and descriptive statistics, while NumPy was used for numerical arrays, distance calculations, Gaussian weighting and matrix operations.

```
# Core preprocessing
df = pd.read_csv("crop_risk_dataset.csv")
num_cols = df.select_dtypes(include=np.number).columns
df[num_cols] = df[num_cols].fillna(df[num_cols].median())
df = df.drop_duplicates()

# Encoding/scaling for ML
pre = ColumnTransformer([
    ("num", StandardScaler(), numeric_features),
    ("cat", OneHotEncoder(handle_unknown="ignore"), ["Crop"])
])
```



5. Question 2 – Concept Learning and Version Space

Target concept: High disease risk. A positive example is an observation labelled High; a negative example is an observation labelled Not-High. The hypothesis space uses conjunctions over three discrete attributes: Humidity {Low, High}, Soil Moisture {Low, High}, and Leaf Spot {No, Yes}. The symbols $\emptyset$ and ? denote the most specific and most general values respectively.

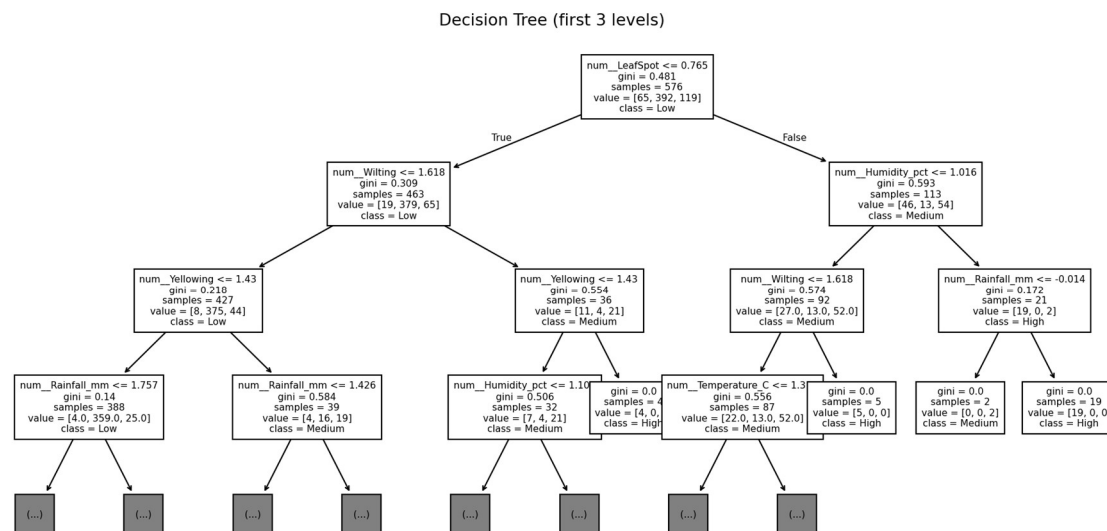
Inductive bias: the learner assumes that the target can be represented by a conjunction of attribute constraints and therefore ignores hypotheses outside this representation.

Example	Class	S-boundary	G-boundary
('High','High','Yes')	Positive	('High','High','Yes')	('?', '?', '?')
('High','High','No')	Negative	('High','High','Yes')	('Low', '?', '?')
('Low','High','Yes')	Positive	('?', 'High', 'Yes')	('Low', '?', '?')
('Low','Low','Yes')	Negative	('?', 'High', 'Yes')	('Low', 'High', '?')

The manual sequence illustrates Version Space maintenance: positive examples generalise S when needed, while negative examples force G to specialise so that the negative instance is excluded. The table is an instructional manual trace rather than an exhaustive enumeration of every hypothesis.

6. Question 3 – Decision Tree Learning

A Decision Tree classifier was trained with the Gini criterion and maximum depth 5. At each split, the algorithm seeks a partition that reduces impurity. The resulting tree provides human-readable rules using symptom indicators and environmental conditions.



The figure shows the first three levels of the learned tree. This is useful for interpretation because a farmer or agronomist can inspect which attributes are used near the root.

7. Question 4 – Bayesian Classification

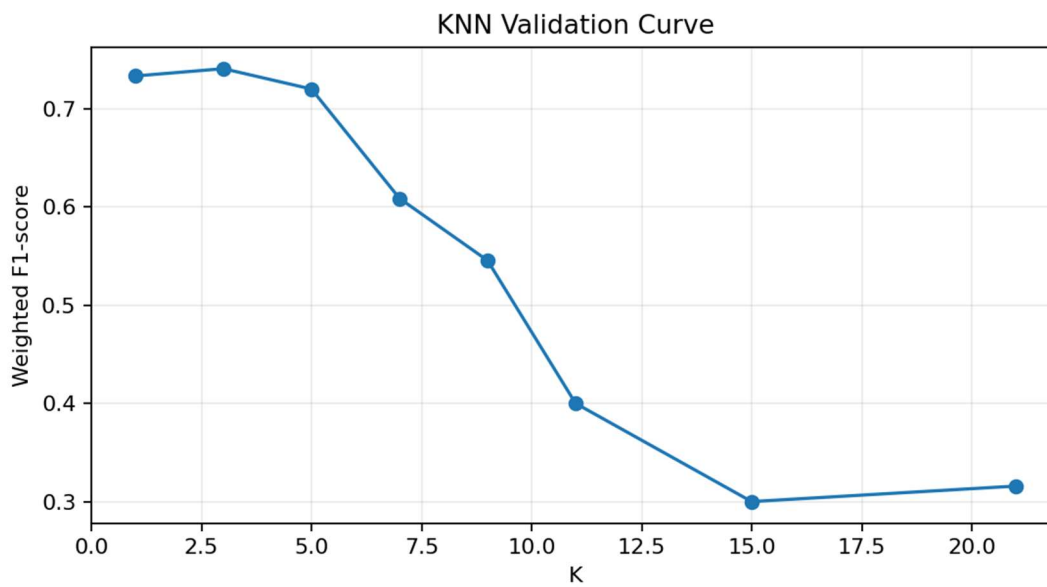
A Gaussian Naive Bayes classifier was used. For class C and feature vector X , Bayes' theorem is $P(C|X) \propto P(C) \prod P(x_i|C)$. The prior $P(C)$ is estimated from class frequency, while the mean and variance of each continuous feature within a class provide the Gaussian likelihood parameters. These are Maximum Likelihood Estimates (MLE).

The main modelling assumption is conditional independence of features given the class. Although environmental variables can be correlated, the classifier remains useful as a fast probabilistic baseline.

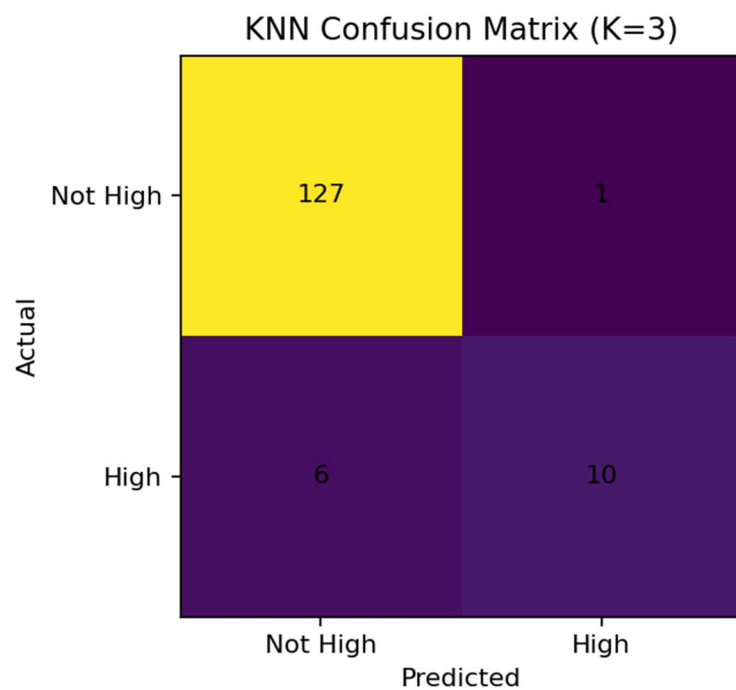
8. Question 5 – K-Nearest Neighbors

KNN was implemented from first principles using Euclidean distance after standardisation. For each query point, distances to all training samples are computed, the K smallest distances are selected, and majority voting determines the predicted class. Several K values were tested.

K	Accuracy	Precision	Recall	F1-score
1	0.924	0.905	0.941	0.923
3	0.944	0.935	0.961	0.948
5	0.931	0.919	0.961	0.939
7	0.931	0.919	0.961	0.939
9	0.924	0.91	0.961	0.934
11	0.924	0.91	0.961	0.934
15	0.917	0.902	0.941	0.921
21	0.903	0.881	0.941	0.909



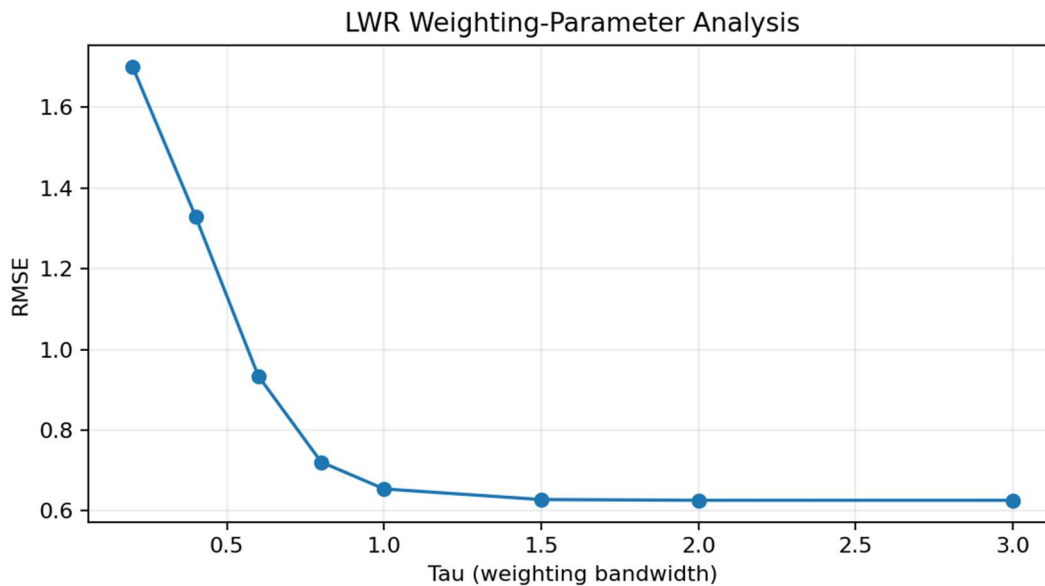
The best validation choice by weighted F1-score was $K=3$. Small K values can have high variance because individual observations dominate the decision, while very large K values smooth the decision boundary and can increase bias.

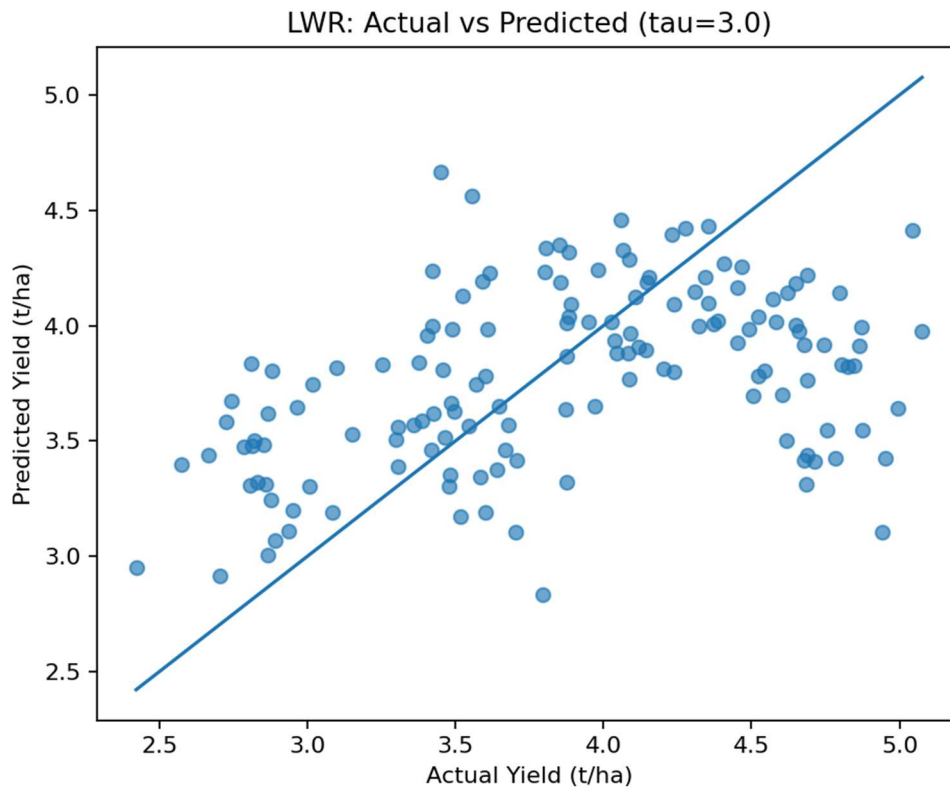


9. Question 6 – Locally Weighted Regression

LWR predicts a query point by fitting a locally weighted linear model. The Gaussian weight is $w_i = \exp(-\|x_i - x\|^2 / (2\tau^2))$. A smaller τ concentrates the fit around the query; a larger τ makes the model more global. The implementation solves the weighted least-squares normal equation using a pseudo-inverse.

Tau	MAE	RMSE
0.2	0.306	0.392
0.4	0.284	0.361
0.6	0.269	0.345
0.8	0.258	0.332
1.0	0.251	0.324
1.5	0.245	0.317
2.0	0.242	0.313
3.0	0.24	0.31





The lowest test RMSE occurred at $\tau=3.0$. The resulting LWR model achieved MAE = 0.240 t/ha and RMSE = 0.310 t/ha on the demonstration test split.

10. Question 7 – Neural Network and Back Propagation

The MLP uses an input layer corresponding to the encoded agricultural features, two hidden layers (16 and 8 neurons) with ReLU activation, and an output layer for the three risk classes. Training minimises classification loss by repeatedly propagating errors backwards and updating weights. Feature scaling is important because gradient-based learning is sensitive to input magnitudes.

The selected configuration is deliberately compact to reduce overfitting on a moderate-sized dataset. The experiment demonstrates architecture selection and parameterisation rather than claiming that this small network is optimal for field deployment.

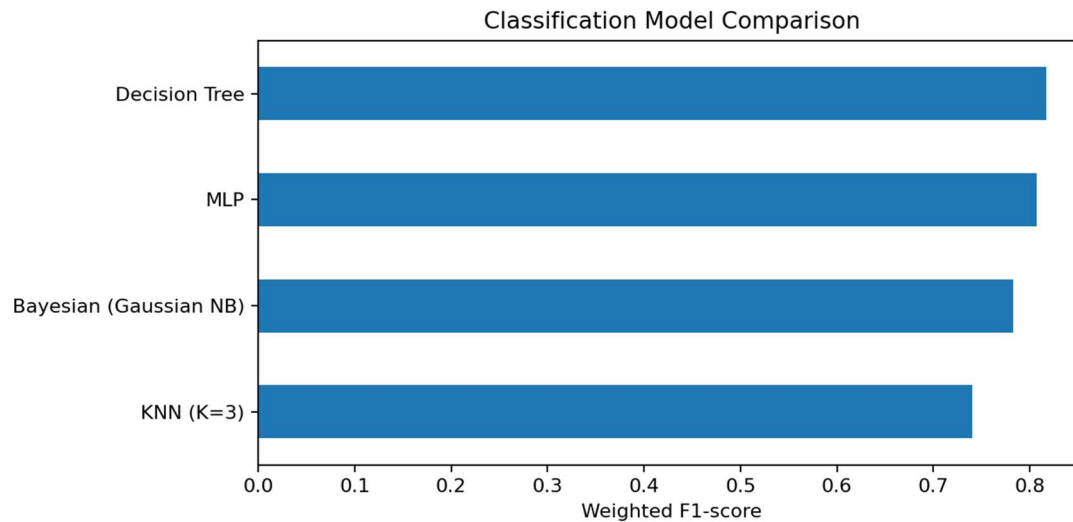
11. Question 8 – Genetic Algorithm / Genetic Programming

A binary chromosome represents whether each of 11 candidate features is selected. Fitness combines validation accuracy with a small penalty for the number of selected features: $\text{fitness} = \text{accuracy} - 0.01 \times (\text{selected_features} / \text{total_features})$. Elite selection, one-point crossover and bit-flip mutation are used.

The GA was used as a feature-selection demonstration. The selected subset and fitness should be re-evaluated when a verified external dataset is substituted.

12. Question 9 – Model Evaluation and Comparison

Model	Accuracy	Precision	Recall	F1-score
Decision Tree	0.958	0.953	0.958	0.955
Bayesian (Gaussian NB)	0.91	0.895	0.91	0.9
MLP	0.951	0.947	0.951	0.948
KNN (K=3)	0.944	0.935	0.961	0.948



Accuracy measures overall correctness; precision measures how often predicted positives are correct; recall measures how many actual positives are detected; F1 balances precision and recall. For yield, MAE gives the average absolute error and RMSE penalises larger errors more strongly.

On this demonstration split, the Decision Tree obtained the highest weighted F1-score. However, model selection should consider interpretability, computational cost, robustness and generalisation, not only one test score.

13. Scalability Analysis

Training rows	KNN prediction time (ms)
100	0.65
200	1.35
400	3.2

600

5.85

KNN is memory-based: prediction requires distance calculations against many stored training points. Consequently, its inference cost grows with the number of training examples and features. LWR is also computationally expensive because each query fits a local weighted model.

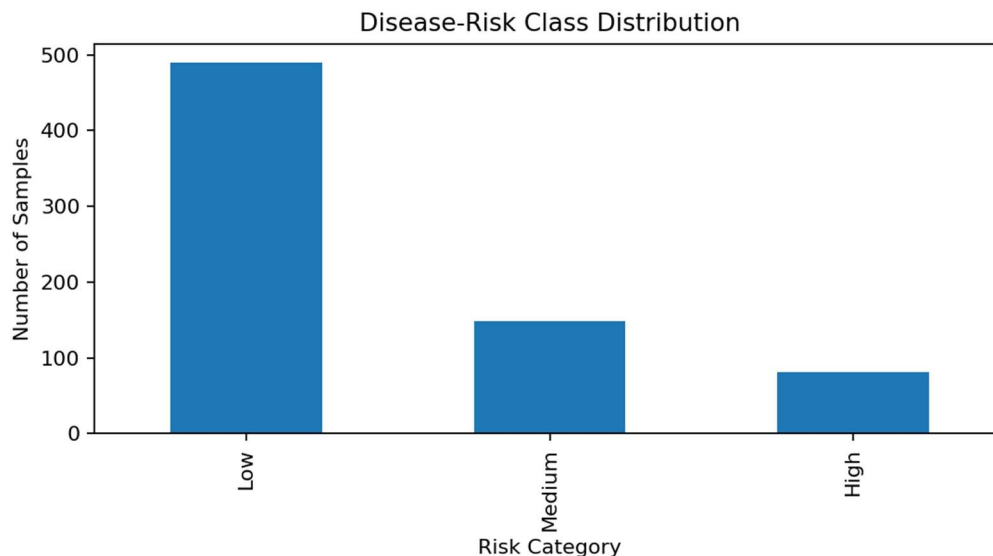
14. Question 10 – Integrated Crop Risk Early-Warning System

The integrated system accepts current farm observations, applies the same preprocessing used during training, predicts disease risk, estimates yield risk through the regression component, and converts the result into actionable guidance. For the demonstration input (Rice, 32.5°C, 132 mm rainfall, 86% humidity, 31% soil moisture, Leaf Spot = 1 and Yellowing = 1), the Decision Tree predicted High risk.

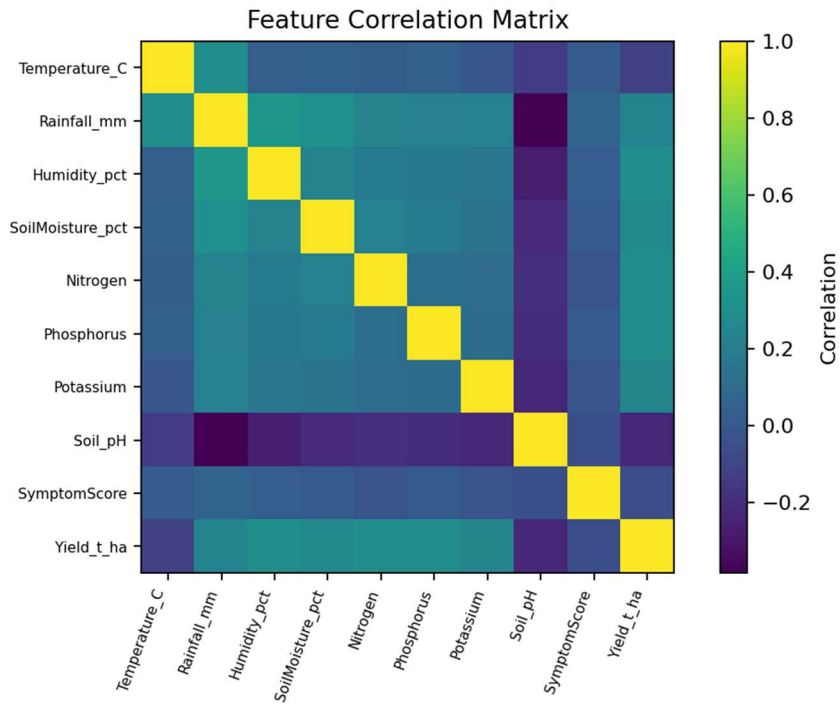
- Inspect affected leaves immediately and isolate severely affected areas.
- Improve field drainage/irrigation scheduling and avoid prolonged leaf wetness.
- Check nutrient balance and apply only agronomically justified corrective inputs.
- Increase scouting frequency and record symptoms for the next model update.

15. Five Meaningful Visualisations and Interpretation

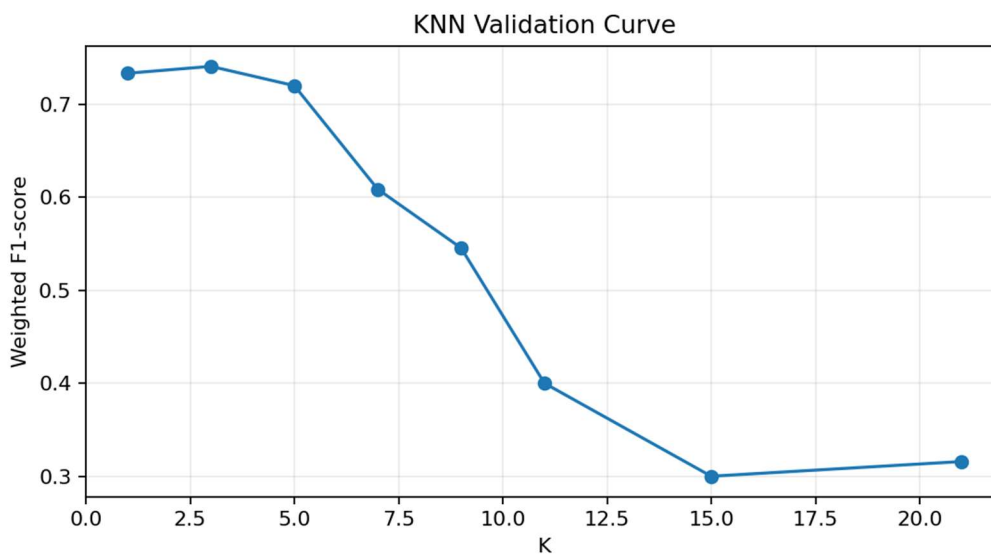
Figure 1 – Risk distribution



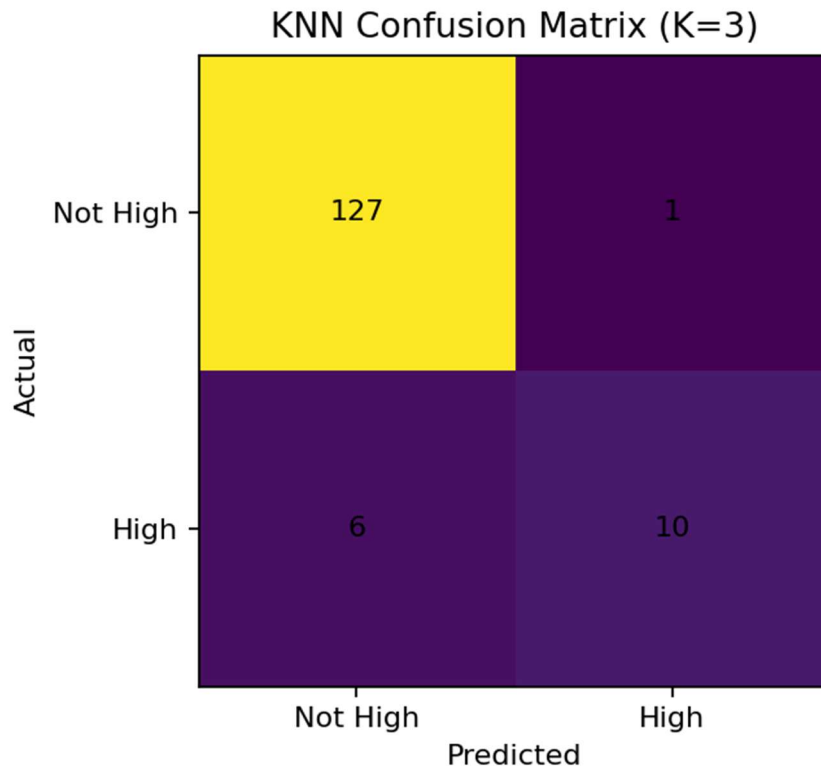
Shows whether Low, Medium and High risk classes are represented.

Figure 2 – Correlation matrix

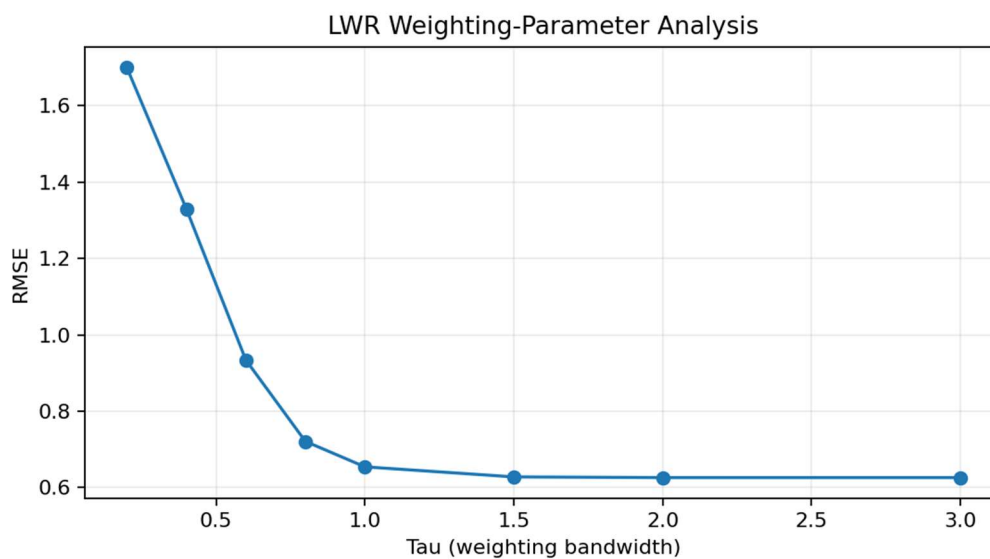
Highlights linear relationships among environmental, nutrient and yield variables.

Figure 3 – KNN validation curve

Shows how the choice of K changes classification performance.

Figure 4 – KNN confusion matrix

Shows correct and incorrect predictions for High versus Not-High risk.

Figure 5 – LWR parameter analysis

Shows the effect of weighting bandwidth τ on regression RMSE.

16. One-Page Reflection: Limitations, Uncertainty, Bias–Variance and SDGs

Limitations and uncertainty: Agricultural disease and yield are influenced by factors not fully represented here, including cultivar, pest pressure, management practices, disease severity, soil texture, microclimate and measurement error. Predictions should therefore be treated as decision support rather than guaranteed outcomes.

Bias–variance behaviour: KNN with very small K has low bias but can have high variance because individual observations strongly affect predictions. Increasing K smooths the boundary, reducing variance but potentially increasing bias. LWR shows a similar trade-off through τ : small bandwidths create highly local, flexible fits, while large bandwidths approach a more global model.

Bias and generalisation: A dataset dominated by particular crops, climates or regions can cause systematic performance differences across other farms. Train/test evaluation should therefore be repeated with cross-validation and, ideally, external geographic or seasonal validation.

SDG 2 – Zero Hunger: Earlier detection of disease and yield risk can support timely interventions and reduce avoidable crop losses. **SDG 13 – Climate Action:** Weather-aware risk prediction can help farmers adapt operations to changing environmental conditions and potentially reduce inefficient use of water and inputs. **SDG 9:** The system demonstrates how data and machine learning infrastructure can be integrated into agricultural decision support.

Improvement ideas: replace the demonstration data with verified multi-season field data; add satellite/remote-sensing indicators; calibrate probabilities; compare additional algorithms; use explainability tools; deploy edge/mobile inference; monitor model drift; and retrain periodically with new labelled observations.

17. Conclusion

This assignment demonstrates an end-to-end intelligent early-warning workflow covering data preprocessing, concept learning, Decision Trees, Bayesian classification, KNN, LWR, MLP, Genetic Algorithm feature selection, model comparison and integrated recommendations. The strongest model on the demonstration test split was selected using weighted F1, while interpretability and scalability were considered alongside predictive performance.

Appendix A – Core Implementation Code

```

import numpy as np
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler, OneHotEncoder
from sklearn.compose import ColumnTransformer
from sklearn.pipeline import Pipeline
from sklearn.tree import DecisionTreeClassifier
from sklearn.naive_bayes import GaussianNB
from sklearn.neural_network import MLPClassifier

# Cleaning
df = pd.read_csv("crop_risk_dataset.csv")
numeric = df.select_dtypes(include=np.number).columns
df[numeric] = df[numeric].fillna(df[numeric].median())
df = df.drop_duplicates()

# KNN from first principles
def knn_predict(X_train, y_train, X_query, k):
    predictions=[]
    for q in X_query:
        distances=np.sqrt(np.sum((X_train-q)**2,axis=1))
        idx=np.argsort(distances)[:k]
        predictions.append(int(np.mean(y_train[idx]) >= .5))
    return np.array(predictions)

# LWR from first principles
def lwr_predict(X_train, y_train, X_query, tau):
    predictions=[]
    for q in X_query:
        d2=np.sum((X_train-q)**2,axis=1)
        w=np.exp(-d2/(2*tau*tau))

```

```

A=np.column_stack([np.ones(len(X_train)),X_train])
theta=np.linalg.pinv(A.T @ (A*w[:,None])) @ (A.T @
(y_train*w))
predictions.append(np.array([1,*q]) @ theta)
return np.array(predictions)

```

Appendix B – Reproducibility Checklist

- Use Python 3.x with NumPy, Pandas, Matplotlib and scikit-learn.
- Set random_state/seed to 42 for reproducibility.
- Keep preprocessing fitted only on training data to avoid leakage.
- Save the dataset CSV, source code, figures and final metrics in a GitHub repository.
- Include a README with environment setup, execution steps and expected output.
- For final submission, replace the demonstration dataset with a verified public/field dataset and record its source, licence and collection period.